

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B. Tech. Branch : Mechanical Semester :V Subject Code & Name: BTMOE505B Renewable Energy Sources Max Marks: 60 Date:19/08/2023 Duration: 3 Hr.			
<i>Instructions to the Students:</i> 1. <i>All the questions are compulsory.</i> 2. <i>The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.</i> 3. <i>Use of non-programmable scientific calculators is allowed.</i> 4. <i>Assume suitable data wherever necessary and mention it clearly.</i>			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the difference between renewable and non-renewable energy with suitable examples. Why is renewable energy important?		6
B)	Discuss in short the different energy sources in India. What is the largest source of energy in India?		6
C)	What is nuclear fusion and fission? What is the present status of nuclear energy in India?		6
Q.2	Solve Any Two of the following.		12
A)	Explain the following terms with reference to solar geometry: (a) Solar altitude angle, (b) Zenith angle, and (c) Air mass		6
B)	Explain the following terms: (a) Direct solar radiation, (b) Diffuse solar radiation, and (c) Reflected solar radiation		6
C)	What is attenuation of solar radiation in the atmosphere? Discuss in short different forms of attenuation of solar radiation in the atmosphere?		6
Q. 3	Solve Any Two of the following.		12
A)	Draw a simple sketch of a flat plate collector and briefly describe the function of each component.		6
B)	What are the different types of concentrating type solar collectors? Describe any one in short.		6
C)	What is the principle of working of a evacuated tube collector? Whether you will prefer flat plate collector or evaluated tube collector for domestic water heating applications? Why?		6

Q.4	Solve Any Two of the following.		12
A)	Explain the following terms with reference to wind power plants: (a) Cut-in speed, (b) Cut-out speed, (c) rated speed, (d) Betz limit (e) Tip speed ratio and (e) wind farm.		6
B)	What is the principle of photovoltaic conversion of solar energy? Explain in short two applications of photovoltaic systems.		6
C)	What is the basic principle of solar still? With a simple sketch explain the working of a solar still.		6
Q. 5	Solve Any Two of the following.		12
A)	What is the difference between pyrolysis, combustion, and gasification, of a biomass?		6
B)	What is the open and closed cycle of OTEC plant? With simple sketch explain working of a closed cycle OTEC plant.		6
C)	What are the disadvantage of a large scale hydropower? What are mini and micro hydropower plants?		6
	*** End ***		

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